

Badr ALKHAMISSI

PhD Candidate at EPFL

@ badr.alkhamissi@epfl.ch +41767603934

https://bkhmsi.github.io github.com/bkhmsi linkedin.com/in/bkhmsi scholar.google.com/bkhmsi

I am passionate about building intelligent systems that can solve complex machine learning tasks and at the same time reveal insights about the inner workings of biological agents. Such systems do not only pave a way for solving long-standing societal problems but also shed light on what it means to be human. Please see [website](#) for a more detailed resumé.

EDUCATION

Present September 2023	EPFL, PHD, Lausanne, Switzerland > Co-advised by Antoine Bosselut (NLP Lab) and Martin Schrimpf (NeuroAI Lab).
September 2020 September 2019	Goldsmiths, University of London, MASTER OF SCIENCE, London, UK > MSc. in Computational Cognitive Neuroscience - Pass with Distinction. > Thesis: Episodic Meta-Reinforcement Learning on the Two-Step and Harlow Tasks (Grade: 90%).
June 2018 September 2014	The American University in Cairo, BACHELOR OF SCIENCE, Cairo, Egypt > BSc. in Computer Science - Summa Cum Laude. GPA 4.0 > Thesis: Video Search via Natural Language.

SELECTED RESEARCH AND INDUSTRIAL EXPERIENCE

November 2022 November 2021	AI Resident, META AI, Seattle, US > Research in NLP. Published 5 first-author papers, and 1 co-author paper. (See Publications) NLP Hate Speech Responsible AI Reasoning Arabic NLP FAIR
October 2021 October 2020	Research Intern, SONY AI, Tokyo, Japan (Remotely) > Published 2 papers in collaboration with DeepMind and 1 paper with Sony CSL. (See Publications) Meta-learning Reinforcement Learning Symbolic Alchemy
September 2019 April 2019	Applied Scientist, MICROSOFT ADVANCED TECHNOLOGY LAB IN CAIRO, Cairo, Egypt > Part of the LUIS Applied Science Team and the Machine Teaching Innovation Group. NLU NLP MT Keras Tensorflow PyTorch Python C#
March 2019 January 2018	Machine Intelligence Researcher, VKANSEE, Cairo, Egypt > Research in fingerprint matching, spoofing and quality analysis through deep learning. Computer Vision Few-Shot Learning Keras Tensorflow Python C++ QT

SELECTED PUBLICATIONS

Selected Preprints

APTERTUS: DEMOCRATIZING OPEN AND COMPLIANT LLMs FOR GLOBAL LANGUAGE ENVIRONMENTS Paper Media Models Tweets Authors: Alejandro Hernández-Cano, Alexander Hägele, [...], Badr AlKhamissi , [...], Antoine Bosselut*, Imanol Schlag*, Martin Jaggi*	UNDER REVIEW
MEDITRON: OPEN MEDICAL FOUNDATION MODELS ADAPTED FOR CLINICAL PRACTICE Paper github.com/https://github.com/epfLLM/meditron Authors: Zeming Chen, Angelika Romanou*, Antoine Bonnet*, Alejandro Hernández-Cano*, Badr AlKhamissi *, Kyle Matoba*, [...], Physician Evaluation Group, Noémie Boillat-Blanco, Kristina Keitel, Javier Elkin, Blaise Robert, Syrielle Montariol, Mary-Anne Hartley*, Martin Jaggi*, Antoine Bosselut*	PREPRINT 2024
RATIONAL METAREASONING FOR LARGE LANGUAGE MODELS Paper Tweets Authors: C. Nicolò De Sabbata, Theodore R. Sumers, Badr AlKhamissi , Antoine Bosselut, Thomas L. Griffiths	UNDER REVIEW
SHADOW-CAVE MODELS: HOW PLATO'S ALLEGORY ILLUMINATES THE LIMITATIONS OF LARGE LANGUAGE MODELS Paper Tweets Authors: Muhammad ElNokrashy*, Badr AlKhamissi *	PREPRINT 2023
A REVIEW OF LANGUAGE MODELS AS KNOWLEDGE BASES Paper Website Tweets Authors: Badr AlKhamissi *, Millicent Li*, Asli Celikyilmaz, Mona Diab, Marjan Ghazvininejad	PREPRINT 2022

Selected Conferences

- MIXTURE OF COGNITIVE REASONERS: MODULAR REASONING WITH BRAIN-LIKE SPECIALIZATION** ICLR 2026
[Paper](#) [Website](#) [Tweets](#) [github.com/https://github.com/bkhmsi/mixture-of-cognitive-reasoners](https://github.com/bkhmsi/mixture-of-cognitive-reasoners)
Authors: **Badr AlKhamissi**, C. Nicolò De Sabbata, Zeming Chen, Martin Schrimpf*, Antoine Bosselut*
- INDUCING DYSLEXIA IN VISION LANGUAGE MODELS** ICLR 2026
[Paper](#) [Tweets](#) [github.com/https://github.com/epflneuroailab/VWFA-Localization](https://github.com/epflneuroailab/VWFA-Localization)
Authors: Melika Honarmand, Ayati Sharma, **Badr AlKhamissi**, Johannes Mehrer*, Martin Schrimpf*
- HIRE YOUR ANTHROPOLOGIST! RETHINKING CULTURE BENCHMARKS THROUGH AN ANTHROPOLOGICAL LENS** EACL 2026
[Paper](#) [Tweets](#)
Authors: Mai AlKhamissi*, Yunze Xiao*, **Badr AlKhamissi**, Mona Diab
- FROM LANGUAGE TO COGNITION: HOW LLMs OUTGROW THE HUMAN LANGUAGE NETWORK** EMNLP 2025
Oral [Paper](#) [Tweets](#)
Authors: **Badr AlKhamissi**, Greta Tuckute, Yingtian Tang, Taha Binhuraib, Antoine Bosselut*, Martin Schrimpf*
- THE LLM LANGUAGE NETWORK: A NEUROSCIENTIFIC APPROACH FOR IDENTIFYING CAUSALLY IMPORTANT UNITS** NAACL 2025
Oral [Paper](#) [Website](#) [github.com/https://github.com/bkhmsi/llm-localization](https://github.com/bkhmsi/llm-localization)
Authors: **Badr AlKhamissi**, Greta Tuckute, Antoine Bosselut*, Martin Schrimpf*
- TOPOLM: BRAIN-LIKE SPATIO-FUNCTIONAL ORGANIZATION IN A TOPOGRAPHIC LANGUAGE MODEL** ICLR 2025
Oral [Paper](#) [Tweets](#)
Authors: Neil Rathi*, Johannes Mehrer*, **Badr AlKhamissi**, Taha Binhuraib, Nicholas M. Blauch, Martin Schrimpf
- INVESTIGATING CULTURAL ALIGNMENT OF LARGE LANGUAGE MODELS** ACL 2024
[Paper](#) [Tweets](#) [github.com/https://github.com/bkhmsi/cultural-trends](https://github.com/bkhmsi/cultural-trends)
Authors: **Badr AlKhamissi**, Muhammad ElNokrashy, Mai AlKhamissi, Mona Diab
- A CONTEXT-CONTRASTIVE INFERENCE APPROACH TO PARTIAL DIACRITIZATION** ARABICNLP 2024
Oral [Paper](#) [Demo](#) [github.com/https://github.com/munael/ccpd-partial-arabic-diacritization](https://github.com/munael/ccpd-partial-arabic-diacritization)
Authors: Muhammad ElNokrashy, **Badr AlKhamissi**
- INSTRUCTION-TUNING ALIGNS LLMs TO THE HUMAN BRAIN** COLM 2024
[Paper](#) [Tweets](#)
Authors: Khai Loong Aw, Syrielle Montariol*, **Badr AlKhamissi***, Martin Schrimpf, Antoine Bosselut
- DEPTH-WISE ATTENTION (DWATT): A LAYER FUSION METHOD FOR DATA-EFFICIENT CLASSIFICATION** LREC-COLING 2024
Oral [Paper](#) [Poster](#)
Authors: Muhammad ElNokrashy*, **Badr AlKhamissi***, Mona Diab
- “FLEX TAPE CAN’T FIX THAT”: BIAS AND MISINFORMATION IN EDITED LANGUAGE MODELS** EMNLP 2024
[Paper](#)
Authors: Karina Halevy, Anna Sotnikova, **Badr AlKhamissi**, Syrielle Montariol, Antoine Bosselut
- ALERT: ADAPT LANGUAGE MODELS TO REASONING TASKS** ACL 2023
[Paper](#) [Poster](#)
Authors: Ping Yu, Tianlu Wang, Olga Golovneva, **Badr AlKhamissi**, Siddharth Verma, Zhijing Jin, Gargi Ghosh, Mona Diab, Asli Celikyilmaz
- TOKEN: TASK DECOMPOSITION AND KNOWLEDGE INFUSION FOR FEW-SHOT HATE SPEECH DETECTION** EMNLP 2022
Oral [Paper](#) [Tweets](#)
Authors: **Badr AlKhamissi***, Faisal Ladhak*, Sridi Iyer, Ves Stoyanov, Zornitsa Kozareva, Xian Li, Pascale Fung, Lambert Mathias, Asli Celikyilmaz, Mona Diab

Selected Workshops

- A SELF-SYNTHESIS APPROACH FOR TRAINING VISION-LANGUAGE MODELS WITH DEVELOPMENTALLY PLAUSIBLE DATA BABYLM 2024**
CoNLL 2024 [Paper](#)
Authors: **Badr AlKhamissi***, Yingtian Tang*, Abdülkadir Gökce*, Johannes Mehrer, Martin Schrimpf
- KHATTAT: ENHANCING READABILITY AND CONCEPT REPRESENTATION OF SEMANTIC TYPOGRAPHY** AI4VA 2024
ECCV 2024 [Paper](#) [Website](#) [Tweets](#)
Authors: Ahmed Hussein*, Alaa Elsetohy*, Sama Hadhoud*, Tameem Bakr*, Yasser Rohaim*, **Badr AlKhamissi**
- OPT-R: EXPLORING THE ROLE OF EXPLANATIONS IN FINETUNING AND PROMPTING FOR REASONING SKILLS OF LLMs** NLRSE 2023
ACL 2023 [Paper](#) [Poster](#) [Tweets](#)
Authors: **Badr AlKhamissi**, Siddharth Verma, Ping Yu, Zhijing Jin, Asli Celikyilmaz, Mona Diab
- MULTITASK LEARNING WITH SELF-CORRECTION FOR HATE SPEECH CLASSIFICATION** OSACT5 2022
LREC 2022 **Oral** [Paper](#) [Best Paper Award](#) [Tweets](#)
Authors: **Badr AlKhamissi***, Mona Diab

AutoML 2022 [Paper](#) [Poster](#) [github.com/https://github.com/BKHMSI/alchemist](https://github.com/BKHMSI/alchemist)
 Authors: Badr AlKhamissi, Akshay Srinivasan, Zeb Kurth-Nelson, Sam Ritter

THE EMERGENCE OF ABSTRACT AND EPISODIC NEURONS IN EPISODIC META-RL

MEMARI 2022

NeurIPS 2022 [Spotlight](#) [Oral](#) [Paper](#) [Poster](#)
 Authors: Badr AlKhamissi, Muhammad Elnokrashy, Michael Spranger

SELECTED AWARDS

- > KAUST Rising Stars in AI Recipient. '26
- > Swiss AI Initiative PhD Fellowship Recipient. '25
- > EPFL Doc.Mobility Grant Recipient. '25
- > President's Cup and Mohamed El Beledy Academic Award for class valedictorian - American University in Cairo '18.
- > Mohammed Bin Abdulkarim A. Allehedan Award for highest performing STEM student - American University in Cairo '18.
- > First place on the two categories of the Arabic Reverse Dictionary shared task at ArabicNLP 2023. '23
- > Best Paper Award for the Fine-Grained Hate Speech Detection shared task in Arabic. '22
- > First place in all four categories of the second NADI shared task. '21
- > Best Mentor Award at Microsoft Advanced Technology Lab in Cairo. '19
- > Shortlisted in Dell EMC's Graduation Project Competition - American University in Cairo '18
- > Computer Science Exemplary Student - American University in Cairo '18
- > Outstanding Performance Award in Deep Learning Summer School at Tsinghua University. '17

TEACHING EXPERIENCE

Spring '24,'25,'26 & Fall '24	Teaching Assistant, EPFL, Lausanne, Switzerland > Teaching assistant for Modern Natural Language Processing and Linear Algebra. Python PyTorch NLP LLMs RLHF
Fall '15 & '17 & Spring '16	Undergraduate Teaching Assistant, THE AMERICAN UNIVERSITY IN CAIRO, Cairo, Egypt > Teaching assistant for Deep Learning and Programming Fundamentals courses. C++ Python Tensorflow Keras SFML

SELECTED INVITED TALKS

- > IndabaX Sudan Featured Speaker: Building Responsible & Ethical LLMs.
- > LauzHack Deep Learning Bootcamp: Brain-Inspired Large Language Models.
- > REST-CL Retreat for Students in Computational Linguistics (UPF University) The LLM Language Network.
- > Psychology Days at Riga Stradins University: Vision and Language in Brains, Minds and Machines.
- > Nancy Kanwisher's Lab at MIT: Mixture of Cognitive Reasoners.
- > NCCR Evolving Language Group: Brain Alignment of Untrained Models.
- > Cohere For AI Community Talk (Geo Regional Africa Group): Investigating Cultural Alignment of LLMs.
- > ARBML Board Talk: Investigating Cultural Alignment of LLMs.
- > AiBelmasry Podcast (Guest Speaker): Analyzing LLM's Ability in Reasoning.
- > TEDxMIU Talk: Representation Matters: Celebrating Egyptians in the AI Revolution.
- > Cohere For AI Community Talk: Presenting OPT-R Paper.
- > Mada Masr Podcast Guest Speaker (April 6th 2023 Episode): A New World Shaped by Artificial Stupidity.
- > IndabaX Egypt 2019 Jointly led a tutorial on cross-lingual alignment.

SERVICE

ORGANIZING COMMITTEE

ICLR 2026 Workshop on Representational Alignment (Re-Align). ArabicNLP 2025 (Publicity Chair). ICML Workshop on Large Languages and Cognition (2024). ArabicNLP 2024 (Social Chair).

SELECTED PROGRAM COMMITTEE MEMBER (REVIEWER)

EACL 2025. NeurIPS 2025. EMNLP 2025. ACL 2025. NAACL 2025 (Area Chair). EMNLP 2024. NeurIPS 2024. ArabicNLP 2024. Cognitive Computational Neuroscience (CCN) 2024. ACL 2024. NAACL 2024. EMNLP 2023. ACL 2023. ENLSP-II Workshop at NeurIPS 2022. EMNLP 2022. Main Track and Late-Breaking Workshop at AutoML 2022. I (Still) Can't Believe It's Not Better! Workshop at NeurIPS 2021. Meta-Learn Workshop at NeurIPS 2021, 2020. Learning to Learn Workshop at ICLR 2021.